GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Independent Set 6.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A tablet screen is 5 inches wide and 8 inches tall. How long is its diagonal, to the nearest tenth of an inch?

About 9.4 in

25 + 64 = 89; √89 ≈ 9.43.

2. A fire truck's 50-ft ladder reaches 48 ft up a building. How far from the building is its foot?

14 ft

2,500 − 2,304 = 196; the ladder is c.

3. A ramp rises 3 ft over a run of 14 ft. How long is the ramp, to the nearest tenth of a foot?

About 14.3 ft

9 + 196 = 205; √205 ≈ 14.32.

4. A park is 7 m by 11 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides? Give it to the nearest tenth of a meter.

About 5.0 m shorter

Along two sides: 18 m. Across: √170 ≈ 13.04 m. 18 − 13.04 = 4.96, so 5.0.

5. Will a 10-in pencil lie flat in a box 7 inches by 8 inches, corner to corner?

Yes — the diagonal is about 10.6 in.

49 + 64 = 113; √113 ≈ 10.63 > 10.

6. A kite flies at the end of 25 m of string, straight above a spot 20 m from where the string is held. How high is the kite?

15 m

625 − 400 = 225; the string is c.